

The Program in Action

When you run *guess.py*, the output will look like this:

```
Guess the Number, by Al Sweigart al@inventwithpython.com
```

```
I am thinking of a number between 1 and 100.
You have 10 guesses left. Take a guess.
> 50
Your guess is too high.
You have 9 guesses left. Take a guess.
> 25
Your guess is too low.
--snip--
You have 5 guesses left. Take a guess.
> 42
Yay! You guessed my number!
```

How It Works

Guess the Number uses several basic programming concepts: loops, if-else statements, functions, method calls, and random numbers. Python's `random` module generates pseudorandom numbers—numbers that look random but are technically predictable. Pseudorandom numbers are easier for computers to generate than truly random numbers, and they're considered “random enough” for applications such as video games and some scientific simulations.

Python's `random` module produces pseudorandom numbers from a seed value, and each stream of pseudorandom numbers generated from the same seed will be the same. For example, enter the following into the interactive shell:

```
>>> import random
>>> random.seed(42)
>>> random.randint(1, 10); random.randint(1, 10); random.randint(1, 10)
2
1
5
```

If you restart the interactive shell and run this code again, it produces the same pseudorandom numbers: 2, 1, 5. The video game *Minecraft* generates its pseudorandom virtual worlds from a starting seed value, which is why different players can re-create the same world by using the same seed.

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1. `"""Guess the Number, by Al Sweigart al@inventwithpython.com`
 2. `Try to guess the secret number based on hints.`
 3. `View this code at https://nostarch.com/big-book-small-python-projects`
 4. `Tags: tiny, beginner, game"""`
 - 5.
 6. `import random`
 - 7.